

Errors reveal the depth of human reasoning

Tamal Maharaj ^{1*} and Kenneth W. Regan ²

¹Ramakrishna Mission Vivekananda Educational and Research
Institute, Belur Math, West Bengal, India.

²Department of Computer Science and Engineering, University at
Buffalo (SUNY), Buffalo, New York, USA.

*Corresponding author(s). E-mail(s): tamal@gm.rkmvu.ac.in;
Contributing authors: regan@buffalo.edu;

Abstract

How deeply people think before choosing is the hidden quantity at the centre of bounded rationality, yet rarely measured in natural behaviour. We read it from the structure of human mistakes: across more than a million skill-graded chess decisions scored by an engine at every search depth, the moves of strong and weak players look comparably good to a shallow analysis and separate only as it deepens—what divides them is the depth at which their errors are exposed, the depth of satisficing. It rises with skill in slow play, contracts as the clock tightens, and vanishes in the fastest games. Estimated without timing information, it tracks how long players thought, recovers the known depth of synthetic agents, and improves move prediction only where deep reasoning has room to matter. Every central result survived a prospective replication registered before its data existed. What people get wrong, more than what they get right, reveals how they think.

Keywords: bounded rationality, satisficing, depth of reasoning, decision-making, expertise, chess

Author note.

Results report held-out analyses on the full dataset; strata and the primary interaction were fixed before held-out evaluation. Evaluation is split by player throughout. T. Maharaj formerly published as T. T. Biswas.

047 1 Introduction

048
049 Sixty years of decision science rest on a quantity nobody can see. When Simon pro-
050 posed that people *satisfice*—search until an option clears an acceptability threshold,
051 then stop—he made the depth of that search the load-bearing variable of bounded
052 rationality [1, 2]. The same hidden quantity does the work in the fast-and-frugal pro-
053 gramme [3], in resource-rational analysis, which treats cognition as the optimal use
054 of limited computation [4–7], in heuristics-and-biases accounts of judgement [8, 9],
055 and in the level- k and cognitive-hierarchy models of strategic thinking [10–16]. Yet in
056 practice, how far a person looks before committing has been inferred almost entirely
057 from small staged experiments. A notable recent exception used a controlled board
058 game with behavioural modelling to show that planning depth grows with expertise
059 [17]. Whether such regularities hold in real, repeated, consequential choice—outside
060 the laboratory and at the scale of populations—has been hard even to ask, because
061 natural decisions rarely come with a ground truth for how good each option was, let
062 alone for how good it *looked* at each successive depth of consideration.

063 Chess supplies exactly that, which is why it has served for a century as the model
064 organism of expert cognition [18–22]. Millions of decisions by players of precisely known
065 skill are recorded in real competition, and every one of them can be scored by an engine
066 far stronger than any human [23, 24]—not once, but at every depth of an iterative-
067 deepening search. The engine’s opinion of a move changes as it looks deeper: a move
068 that seems strong at a glance may be refuted only far down the tree, and a move that
069 looks dull at first may reveal its point three moves later. This depth-resolved record
070 lets us put a new question to every human mistake: *at what depth does the flaw become*
071 *visible*? And therefore: how deep must the player have looked?

072 We call the answer the **depth of satisficing**—the depth at which the move a
073 player chose stops looking better than the move they should have played. Our central
074 empirical claim is that this is the dimension along which skill is most legible. Judged
075 only at shallow depth, the moves of players separated by a thousand rating points look
076 comparably good; the gap opens as the engine searches deeper. What divides them
077 is carried by their errors—not by how many they make, but by how deep one must
078 look before the flaw appears. Skill is present even in moves played at a glance, since
079 stronger players recognise better options without searching; but most of what rating
080 buys is realised only through the depth to which a player effectively searches.

081 Online chess also supplies a natural, graded manipulation of how deeply a person
082 can afford to look: the time control. A player with an hour for the game can follow
083 a line many moves ahead; one with ten minutes cannot; and one playing *bullet*—a
084 minute or two for all the moves—has time only to react. Because the same players
085 compete across every control, the clock varies the budget for search while holding the
086 person, and their pattern knowledge, fixed. If the depth of satisficing measures search
087 rather than mere skill, it should track that budget: deepest in classical play, shallower
088 in rapid and blitz, and collapsing in bullet, where even strong players must move on
089 recognition alone with no time to verify what they see. This graded prediction across
090 four controls is among the sharpest tests we can put to the construct, and its limiting
091 case is a built-in control: in bullet, search is unavailable to everyone, so any residual
092 skill difference there is pattern, not depth.

Three things distinguish this work from earlier studies of chess errors and engine depth [25–27]. First, we replace descriptive curve-fitting with a generative model: the player is a latent distribution over search depths, so the depth of satisficing is a posterior quantity with stated uncertainty rather than the crossing point of two fitted curves. Second, we do not assume that engine depth is a valid proxy for human reasoning depth; we test it—predictively, convergently, and differentially. Third, we subjected the construct to an adversarial replication programme: hypotheses, pipeline, and decision rules were pre-registered and then confirmed on untouched months of data, ending with a fully prospective test registered before its data existed [28]. The final, registered confirmation succeeded on all three primary hypotheses.

2 Results

Reading a decision at increasing depth.

A chess engine evaluates by *iterative deepening*: it scores every candidate move looking one ply (half-move) ahead, then two, and so on, revising each move’s evaluation at every new horizon; evaluations change with depth because deeper search uncovers consequences that shallower search cannot see—the game-tree analogue of thinking further ahead. A move’s value is therefore not a number but a *trajectory* across depth, and that trajectory is our raw material. Figure 1a shows a real decision from the data. A 1910-rated player, offered a bishop, recaptured it (fxe6): at shallow depth the capture looks close to winning, but from about eight plies the refutation comes into view and its value collapses. The quiet king move Kb8 shows the opposite signature—unimpressive at a glance, it proves best once the search runs deep. Writing $\delta_{m,d}$ for move m ’s *regret* at depth d —the win probability it concedes relative to the best option visible at that depth (Methods)—we call the signed area between a move’s regret trajectory and its full-depth endpoint the move’s *swing*, $\text{sw}(m) = \sum_d (\delta_{m,d} - \delta_{m,D})$: positive (*swing-up*, a deep discovery) when a move looks worse shallow than it truly is, negative (*swing-down*, a trap) when shallow search flatters it. A position’s total swing, $\sum_m |\text{sw}(m)|$, measures how much the picture changes with depth—how much deliberation the position can reward. The depth at which the played move’s flaw or worth becomes visible is the signal everything below is built on.

Skill lives in the errors, not the successes.

Our primary sample comprises 922,412 decisions by 16,328 players rated 1200 to 2600+, drawn from online play and from elite over-the-board tournaments (Methods). At shallow engine depth, the average error of the played move barely distinguishes a club player from a grandmaster; the gap opens only as the engine looks deeper (Fig. 1b). Because online ratings are pool-specific—a separate rating per time control—we read every skill comparison *within* a control: at full depth, mean regret of the played move falls with rating in classical (0.059 at the weakest band to 0.028 at the strongest), in rapid, and in blitz, but stays nearly flat in bullet (0.070 to 0.055), where no one has time to search deeply. The separating information, moreover, is not spread evenly across mistakes. It is concentrated in *deep-discovery* decisions—swing-up moves whose worth only emerges at depth—which fan out cleanly by rating (Fig. 1c), while

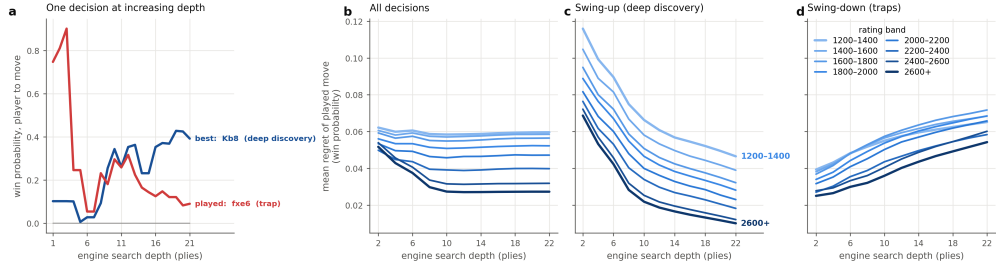


Fig. 1 Skill separates only at depth, and mostly on deep-discovery moves. **a**, A worked example: candidate-move win probabilities across engine search depth for one real decision (grey, other candidates; moves in standard algebraic notation). The played pawn capture fxe6 looks nearly winning at shallow depth and collapses once the refutation surfaces (a *trap*, swing-down); the quiet king move Kb8 reveals its worth only deep (a *deep discovery*, swing-up). **b–d**, Mean regret of the played move across engine search depth, by rating band (light to dark blue: 1200–1400 up to 2600+), within the slow controls (classical and rapid). **b**, All decisions: rating bands are nearly indistinguishable at shallow depth and fan apart as the engine looks deeper. **c**, Swing-up (deep-discovery) decisions: a clean, well-ordered rating fan—the rating signal concentrates here. **d**, Swing-down (trap) decisions: bands bunch and overlap, separating roughly half as much. The rating axis spans online play (lower bands) to elite over-the-board play (upper bands).

trap-like swing-down decisions bunch the rating bands together (Fig. 1d). A single swing-up decision carries roughly twice the rating information of a swing-down one (pool-normalised single-decision $R^2 \approx 0.006$ versus 0.003), an asymmetry that recurs in every replication month below. What matters is not how large an error is, but at what depth the move’s true worth appears.

Depth of satisficing rises with skill and collapses under time pressure.

Fitting the latent-depth model (Methods) and reading off each decision’s posterior expected depth, the construct behaves exactly as a measure of deliberation should—provided, again, that one looks within a time control, since pooling would conflate skill with available clock. Within classical games, the depth of satisficing climbs from about 10.8 plies in the lowest rating band to about 13.7 in the highest, with non-overlapping player-clustered 95% intervals at the extremes (Spearman $\rho = +0.40$; rapid +0.66; blitz +0.45; Fig. 2a). Bullet is the built-in negative control: the relationship vanishes ($\rho = -0.04$, flat across bands). When nobody can search, depth stops separating anybody, which rules out the construct being a re-description of rating. The same signature appears *within* players: de-meaned depth rises across think-time quartiles, from -0.14 plies on a player’s fastest quarter of moves to $+0.33$ on their slowest (Fig. 2b).

What the clock buys.

The four controls form a graded ladder of thinking time rather than arbitrary labels. In the online sample the median time a player actually spends on a move falls from about 11 seconds in classical play to 5 in rapid, 3 in blitz, and 1 in bullet—close to an order of magnitude, tracking base budgets from thirty minutes down to one. This ladder is what the depth of satisficing responds to: with more clock a player can carry a line further before committing, so depth rises with skill in classical, rapid, and blitz,

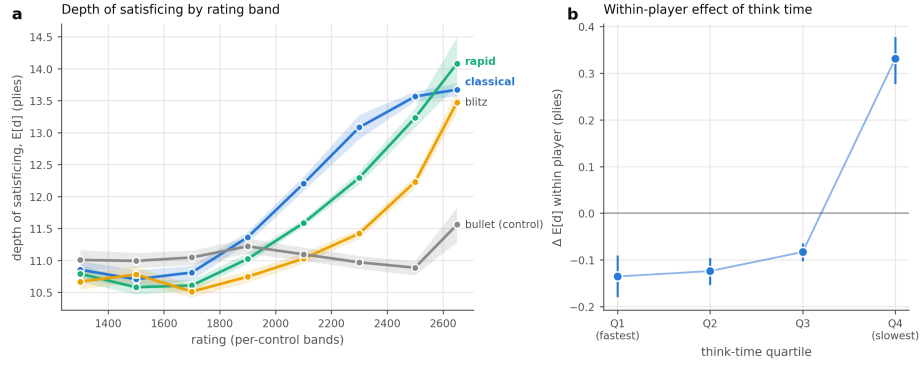


Fig. 2 Depth of satisficing rises with skill and with time to think. **a**, Posterior depth of satisficing by rating band, within each time control (shaded bands, player-clustered bootstrap 95% CIs). Depth rises with rating in classical (Spearman $\rho = +0.40$), rapid (+0.66) and blitz (+0.45); bullet, where no one can search, is the flat negative control ($\rho = -0.04$). **b**, Within-player effect: de-meaned depth by think-time quartile (dots, pooled means; bars, player-clustered bootstrap 95% CIs). The same players search deeper on their slower moves.

where there is time to search, and goes flat in bullet, where a move must be produced in about a second—from recognition rather than search. The within-player pattern says the same thing from the other side: the same person searches deeper on their slower moves (Fig. 2b). The clock sets the ceiling on attainable depth, and skill is expressed in how fully a player uses the room the clock allows—which is why time pressure compresses the skill gap and slow play opens it.

Inferred depth tracks real thinking time it was never shown.

The construct’s key non-circular test removes the clock from the model entirely. Refit with the clock feature zeroed, so that timing can play no role in the estimate, the model still assigns held-out depths that correlate with how long the player actually spent on the move (Spearman $\rho = +0.36$ overall), and the correlation peaks in the middlegame (+0.40)—the phase where deliberation genuinely varies most—falling to +0.33 in the opening and +0.29 in the endgame (Fig. 3). Inferred depth recovers observed deliberation without ever being told about time. That is convergent evidence that the posterior reflects thinking, not the rating prior it partly leans on.

A depth-aware model predicts human moves better, exactly where depth should matter.

If the depth of satisficing is real, knowing it should help predict what a human will play, and the help should be confined to situations where depth can vary and can matter. Both hold. A fusion of a state-only human-move model (Maia-3 [29]) with a mixture-over-depths search model improves held-out cross-entropy over the state-only model by a modest 0.012 nats overall (95% CI [0.008, 0.016], player-clustered bootstrap; top-1 accuracy +0.7 points). The scientific content is not the average but the pre-registered differential: the gain concentrates in classical, high-swing positions (+0.053 nats) and is absent or negative in fast or low-swing play (Fig. 4). The swing-magnitude $\times$ time-control interaction is +0.060 nats (95% CI [0.056, 0.064]), confirmed

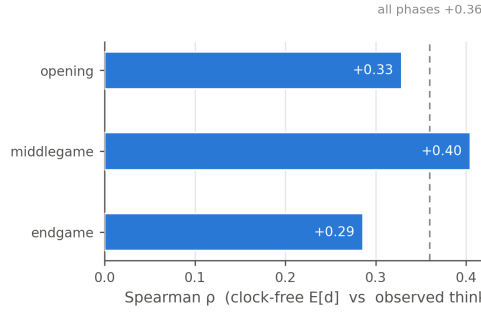


Fig. 3 Depth inferred without the clock tracks real thinking time. Held-out Spearman correlation between the clock-free model’s per-decision depth and observed time-on-move, by game phase (dashed line, all phases pooled). The correlation peaks in the middlegame, where deliberation varies most.

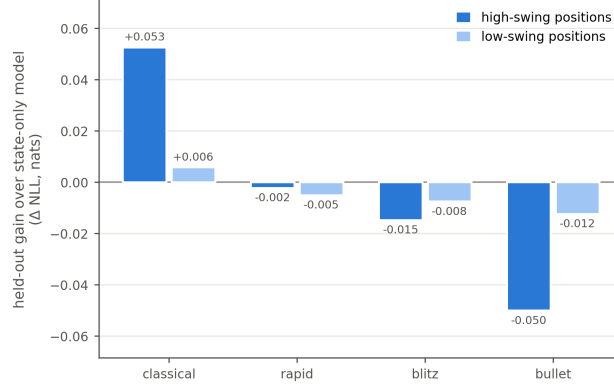


Fig. 4 Depth improves move prediction only where depth can matter. Held-out gain of the depth-aware fusion over the state-only model (nats), by time control and position swing. The gain concentrates in classical, high-swing positions and is absent or negative in fast or low-swing play—the pre-registered differential.

by a mixed-effects model with players as random effects ($\beta = 0.058$, $p < 10^{-277}$). No state-only model can produce this pattern, however strong [30]; within classical play the gain falls almost entirely on swing-up moves (+0.048 nats)—precisely the choices whose worth a pattern model cannot see because it does not look ahead.

Positions are test items; swing is their discrimination.

These findings admit a measurement-theoretic reading that unifies them. Treat each position as a test item whose *difficulty* is its critical depth—the deepest ply at which a misleading move still looks best—and whose graded *response* is the severity of the move played (best, inaccuracy, mistake, blunder). An explanatory graded-response model [31], with item properties entered as engine-derived covariates in the manner of a linear logistic test model [32] (each natural position occurs only once, so per-item parameters are unidentifiable), recovers the full expected structure: error severity falls with ability ($\beta = -0.13$, $p < 10^{-80}$), rises with item difficulty ($\beta = +0.06$, $p < 10^{-24}$),

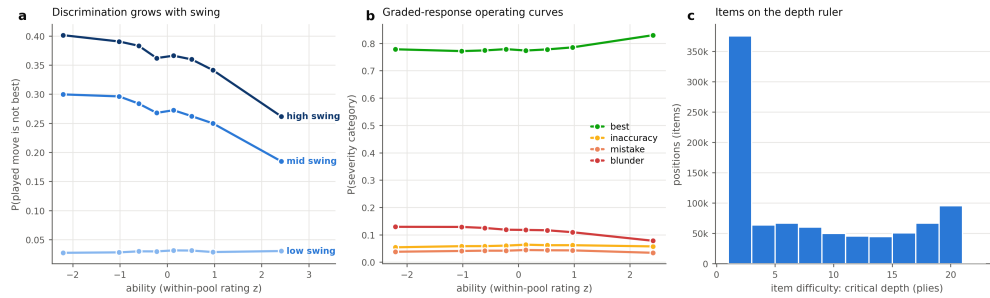


Fig. 5 An item-response reading: swing is item discrimination. **a**, Probability that the played move is not the engine-best, versus ability (within-pool rating z), by position-swing tercile: ability discriminates far more steeply in high-swing positions. **b**, Graded-response operating curves: probability of each severity category versus ability; $P(\text{best})$ rises and $P(\text{blunder})$ falls monotonically. **c**, Items placed on the ply ruler by their critical depth—the same scale on which the depth of satisfying measures person ability.

and—the decisive term—ability discriminates more sharply in high-swing positions (ability $\times$ swing $\beta = -0.08$, $p < 10^{-38}$; Fig. 5a). In item-response language, *swing is item discrimination*: test information concentrates in high-swing decisions, which is exactly why skill is legible in so small a fraction of moves. The graded operating curves shift cleanly along the ability axis (Fig. 5b), and item difficulty lives on the same ply ruler as the person-side depth of satisfying (Fig. 5c)—a natural generalisation of rating systems, which are themselves one-parameter item-response models on game outcomes [33–35]. The structure is not a platform artefact. On an independent pool of 99,358 chess.com decisions by 945 players—a different site, a different rating system—the same model reproduces every sign (ability $\beta = -0.08$, difficulty $+0.11$, swing $+0.87$, ability $\times$ swing -0.06 ; all $p < 10^{-13}$), and the same holds on a second Lichess month (2026-04: ability $\times$ swing $\beta = -0.09$, $p < 10^{-40}$)—three independent pools in all. At the present decision density per player, ability is reliable only in aggregate; person-dense, pool-invariant ability estimation is future work.

Profiles, not a single number, separate expert styles.

Averaged per player from an *elo-free* refit (so that recovering rating is not circular), four behavioural axes—depth of satisfying, trap susceptibility, deep-discovery rate, and time elasticity—summarise a player’s style. Across 873 players with at least 100 decisions, the four-axis profile recovers within-pool rating at $R^2 = 0.10$ under nested cross-validation while the depth axis alone recovers none ($R^2 \approx 0$): the added axes carry the signal, reported on the honest within-pool scale rather than the cross-pool one (which flatters recovery by conflating skill with pool membership). Among elite players the profile also resolves a known embarrassment of one-dimensional scores: a bare depth ranking inverts the true order of the world’s best (9 of 15 elite pairs discordant), ranking, for example, Abdusattorov and Nepomniachtchi above Carlsen. The four axes instead attribute each player’s strength differently—Nakamura is the most trap-resistant of the six yet among the least time-elastic (a blitz specialist who reaches his depth *fast*), while Carlsen converts thinking time into depth more steeply than anyone—and the inversions disappear (Fig. 6).

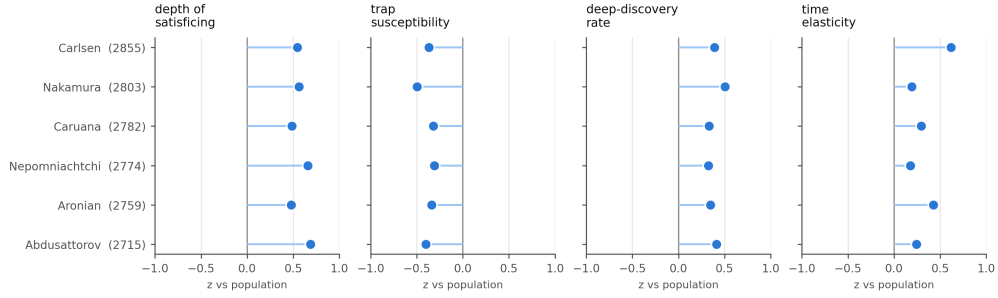


Fig. 6 Four behavioural axes separate elite styles that one number confuses. Profile z -scores (versus the full population) for the six elite players with the most analysed decisions, ordered by rating. A one-dimensional depth score inverts elite ratings (9 of 15 pairs discordant); the four-axis profile recovers within-pool rating across 873 players ($R^2 = 0.10$, nested cross-validation) where depth alone recovers none ($R^2 \approx 0$), and reads the styles directly—for example Nakamura as most trap-resistant but among the least time-elastic, Carlsen as converting think time into depth most steeply.

The instrument recovers depth it has never seen.

Synthetic agents that satisfice at fixed, known depths, played through the identical pipeline, come back in the right order: recovered depth increases monotonically with planted depth (group-level Spearman = 1.0 across planted depths 4–16), and the estimator independently recovers the agents’ decision sharpness ($\hat{\beta} = 7$ against a generating $\beta = 6$; Fig. 7). Recovery is ordinal rather than metric—per-agent estimates are noisy and compressed toward the middle (mean absolute error 3.7 plies on roughly 70 decisions per agent)—but it establishes that the model measures depth rather than position difficulty. Recovery requires a flat prior over depth: imposing the human rating-conditioned prior collapses the estimate toward its mean, which is why the two non-circular tests above, and not the posterior’s absolute scale, carry the evidential weight.

A pre-registered programme ending in a prospective confirmation.

We pre-registered hypotheses, pipeline, and decision rules, and then ran the frozen pipeline once on each of three untouched Lichess months [28]. The first replication (2026-05; 179,360 decisions; OSF KB4ZQ) confirmed the non-circular core—clock-free depth tracked thinking time ($\rho = +0.30$, middlegame peak), and the registered swing $\times$ time-control interaction held ($+0.049$ nats, mixed-effects $p < 10^{-36}$)—but failed its own strict rule because one arm of one primary (blitz, registered as positive) came out flat, and it exposed a swing-label convention error: the single-decision rating signal sits in swing-*up* decisions, not swing-*down* as first registered. We disclosed both, corrected the hypotheses, and registered a second, fully prospective protocol (OSF A6NYK) whose primary dataset—Lichess 2026-06—did not yet exist at registration. A robustness month processed under the same registration (2026-04; 168,654 decisions) meanwhile reproduced the slow-control rating relationship (classical $\rho = +0.47$, rapid $+0.54$), the convergent-time result ($+0.40$), and the interaction ($+0.076$ nats, $p < 10^{-48}$).

The prospective month then arrived and was processed once. All three primary hypotheses replicated, satisfying the registered decision rule. On 203,248 decisions

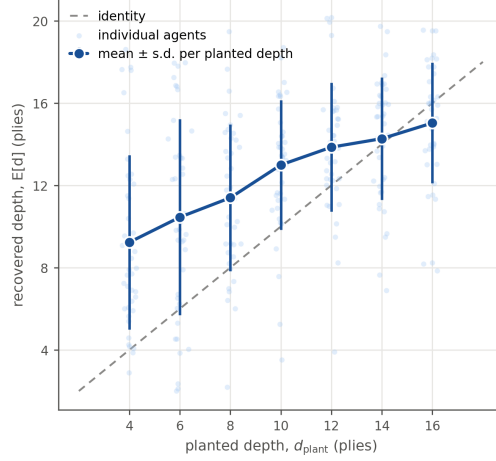


Fig. 7 The instrument recovers planted search depth. Synthetic agents satisficing at fixed depths (4–16 plies), run through the identical pipeline. Pale dots, individual agents; dark points, mean $\pm$ s.d. per planted depth; dashed line, identity. Recovery is monotone (group-level Spearman = 1.0) though ordinal: per-agent estimates are compressed toward the middle (MAE 3.7 plies at ~ 70 decisions per agent).

from 3,707 players: depth rose with rating within every slow control (classical $\rho = +0.23$, 95% CI $[+0.15, +0.35]$; rapid $+0.72$ $[+0.59, +0.79]$) with bullet null as predicted ($+0.08$ $[-0.03, +0.19]$); clock-free depth tracked observed thinking time ($\rho = +0.32$, middlegame peak $+0.36$); and the depth-aware model beat the state-only model overall ($+0.010$ nats, 95% CI $[+0.003, +0.017]$) with the registered interaction intact ($+0.068$ nats $[+0.055, +0.079]$; mixed-effects $\beta = +0.059$, $p < 10^{-42}$). Two secondary outcomes are worth reporting straight. Blitz, which we had over-corrected to “null” after its flat 2026-05 showing, came back clearly positive ($\rho = +0.39$ $[+0.25, +0.50]$); across the four months the blitz relationship now reads $+0.45$, ≈ 0 , $+0.19$, $+0.39$ —real but unstable, in contrast to the robust slow controls and the reliably null bullet. And the deep-discovery asymmetry replicated in its corrected direction (swing-up $R^2 = 0.0049$ versus swing-down 0.0027), as it has in every month examined. The construct’s central claims thus rest on a confirmation that was, in the strict sense, predicted in public before the behaviour occurred.

3 Discussion

Errors are usually the noise of behavioural data. Here they are the signal. Where success is widely shared, the fine structure of failure—the depth at which a flaw becomes visible—turns out to be a stable, measurable signature of how a person reasons: it scales with expertise, contracts under time pressure, tracks deliberation it was never shown, and improves prediction exactly in the regime where deep reasoning has room to operate. The pattern supports a specific mechanism: people commit to the best option found at the search horizon they reach, rather than integrating over the whole trajectory—consistent with theoretical results that level- k predictions are independent of higher-order belief tails [16], and connecting satisficing to model-based control

[36], decision-tree pruning [37], and bounded evidence accumulation [38]. Where planning depth has been measured before, it required controlled laboratory games and explicit search models [7, 17]; here the same construct is read non-invasively from natural, high-stakes play at population scale, against a depth-resolved value standard, and it survives a prospective registered test. The item-response formulation adds a measurement-theoretic dividend: positions become items with engine-derived difficulty and discrimination, so people and problems share one ply-denominated scale—rating systems generalised from game outcomes to individual decisions.

The individual profiles suggest what the construct is for. Expertise is not a scalar: two players of equal rating can owe their strength to depth, to trap resistance, to deep discovery, or to the efficiency with which time becomes depth—echoing evidence that expert flexibility itself varies [39]. An agent that estimates a person’s depth of satisficing could, in principle, select problems pitched just beyond it; a rating system built on decisions rather than outcomes could equate skill across pools that never play each other.

The limitations are the flip side of the design. Engine search depth is a *validated proxy* for human selective search—validated by convergence with thinking time and by recovery of planted depth—but it is not isomorphic to human search, which is narrow, uneven, and pattern-guided. The depth-of-satisficing posterior leans partly on a rating-conditioned prior; per-decision move evidence is weak, which is why the clock-free and synthetic tests, not the posterior’s absolute scale, carry the claims. The average predictive gain is small, and the case rests on its registered differential rather than its size. The blitz rating–depth relationship is genuinely variable across months and should not be quoted as a stable effect. And chess players are a highly trained, self-selected population in an adversarial task; the approach extends in principle to any domain with skill-graded decisions and a depth-resolved value standard—a value network queried at increasing depth in Go, or staged value estimates in planning tasks—but that generality remains to be demonstrated.

443

444

445 4 Methods

446

447 *Data.*

448 The primary dataset comprises 922,412 decisions by 16,328 players: online games from
 449 the Lichess open database (month 2025-09), balanced-sampled across rating bands
 450 (200-point bins, 1200–2600+) and time classes (bullet, blitz, rapid, classical), with per-
 451 move clock times retained. Time classes follow Lichess’s convention, by estimated game
 452 duration (initial time plus $40 \times$ the increment): bullet under 3 minutes, blitz 3–8, rapid
 453 8–25, and classical 25 minutes or more. Each class therefore pools nearby formats—
 454 1+0 and 1+1 bullet, for instance, both fall in the bullet class despite an appreciable
 455 difference in effective time—so residual within-class heterogeneity remains, and every
 456 skill comparison in this paper is made within class, never across; plus an elite over-the-
 457 board stratum of 371,279 decisions from 1,261 titled players in broadcast tournament
 458 games, which supports the individual-profile case study. Replication datasets—Lichess
 459 2026-05 (179,360 decisions), 2026-04 (168,654), and the prospective 2026-06 (203,248
 460 decisions, 3,707 players, 113,421 classical)—were each sampled by the same frozen

pipeline (identical seed, cell targets, and engine settings) and processed once. The independent-pool item-response check uses 99,358 decisions by 945 titled chess.com players harvested via the public Published-Data API. Synthetic agents (below) provide ground truth. All decisions are restricted to plies 9–120 to skip opening-book play.

Engine analysis and value scaling.

Positions are analysed with Stockfish 18 in multi-PV mode, recording each candidate move m_i 's evaluation at every depth $d = 1 \dots D$ ($D=21$, $K=9$). The candidate set is the union of the top- K across all depths, so trap moves keep a full trajectory. Because consecutive plies of each game are analysed, the value of a played move falling outside the top- K is read off from the position it leads to (the next analysed decision): its win probability at depth d equals one minus the successor's best win probability at depth $d-1$, matching the other candidates' negamax depth. Where no analysed successor exists (game end, final sampled ply) the played move takes the weakest retained candidate's win probability at each depth. With UCI_ShowWDL the engine reports win/draw/loss counts (w, dr, ℓ) per mille, giving the win probability $P_{i,d} = (w + \frac{1}{2} dr)/1000$ directly—no logistic or centipawn scaling. Depth- d regret is

$$\delta_{i,d} = \max_j P_{j,d} - P_{i,d} \geq 0. \quad (1)$$

Being a difference against the best move *at each depth*, δ is invariant to anything that shifts all moves together (position difficulty, evaluation scale), isolating the choice problem. Searches run to depth D or a 2×10^8 -node cap, whichever comes first; the 0.09% of positions that stop short have their unreached depths masked in the likelihood (reached depth recorded) rather than imputed.

Pattern prior.

The human-pattern prior $q_i(p, c)$ is the Maia-3 policy [29] at the side-to-move's rating, read from the policy head as a full distribution over legal moves.

Latent-depth choice model.

A player who has effectively searched to depth d chooses by a soft-max over depth- d regret fused with the pattern prior as a product of experts,

$$P(\text{move}=i \mid d, p, c) = \frac{\exp(-\beta \delta_{i,d} + \alpha \log q_i)}{\sum_j \exp(-\beta \delta_{j,d} + \alpha \log q_j)}. \quad (2)$$

Effective depth is latent with distribution $\pi_d(p, c)$, emitted by a small network over a coarse depth grid conditioned on context c (rating, time class, clock, move number, total swing). The move likelihood marginalises depth, $P(\text{move}=i \mid p, c) = \sum_d \pi_d(p, c) P(\text{move}=i \mid d, p, c)$, and parameters minimise the negative log-likelihood of the played move with an entropy penalty on π for identifiability and β tied globally. Data are split by player throughout.

507 ***Depth of satisficing.***

508 The posterior over effective depth given the observed move,

509
510
511
$$r_d = P(d \mid y, p, c) = \frac{\pi_d P(y \mid d)}{\sum_{d'} \pi_{d'} P(y \mid d')}, \quad \hat{d} = \sum_d d r_d, \quad (3)$$

512
513
514 gives a per-decision depth with uncertainty. Lichess ratings are pool-specific (a separate Glicko-2 rating per time control, not comparable across controls), so no analysis
515 pools ratings across controls; doing so would also confound skill with available clock.
516 Per-band estimates carry player-clustered bootstrap 95% intervals (1,000 resamples of
517 players) [40].
518
519

520

521

522 ***Relation to the curve-intersection estimator.***

523 The construct originates in a population-level heuristic [26]: average the played move's
524 error and the engine move's error against depth, and read the depth of satisficing
525 off the curves' crossing. That estimator returns one depth per cohort, carries no per-
526 decision uncertainty, presupposes a single time control, and inherits the fit (cohort
527 depths move by roughly a ply under reasonable fitting choices). We keep the underlying
528 signal—a played move whose error grows with depth is exactly the move *swing* of
529 Biswas and Regan [26]—and replace the crossing with the generative posterior r_d : a
530 per-decision quantity with credible intervals that supports within-control and within-
531 player contrasts and is identifiable against synthetic ground truth. The population
532 curve is stable only because it averages thousands of weak per-move signals, which the
533 crossing view conceals and our identifiability test makes explicit: recovered depth is
534 ordinal and prior-compressed, which is why the convergent and identifiability tests, not
535 the posterior's absolute scale, carry the claims. Where the two views can be compared
536 they agree—stronger players satisfice deeper and err less—now within time control, at
537 far larger scale, on win-probability evaluation, with a human-pattern prior, and under
538 a registered replication programme.

539

540

541 ***Item-response formulation.***

542 Each decision is an item. Difficulty b_j is the deepest grid depth at which the apparent-
543 best move differs from the full-depth best move—the critical depth d_i of Biswas and
544 Regan [26], in plies; the ordered response is the played move's full-depth regret binned
545 as best (≤ 0.02), inaccuracy (≤ 0.05), mistake (≤ 0.10), or blunder (> 0.10) in win-
546 probability units. We fit an explanatory proportional-odds graded-response model
547 [31, 32] with ability (within-pool rating z), b_j , position swing, and ability $\times$ swing
548 as predictors; item properties enter as covariates rather than free parameters because
549 most natural positions occur exactly once. The MLE uses a random subsample;
550 response curves are computed on all decisions. Per-player ability has low split-half
551 reliability at this decision density, so effects are reported at the item and rating-band
552 level.

<i>Profiles and rating recovery.</i>	553
Per player with at least 100 decisions: mean $\hat{d}$; trap susceptibility (rate of played swing-down moves, weighted by shallow attractiveness); deep-discovery rate (rate of played swing-up moves); and time elasticity (within-player slope of $\hat{d}$ on log think-time). The depth axis comes from a model fit with the rating feature removed, keeping recovery non-circular. Rating is predicted from the standardised profile by ridge regression under nested five-fold cross-validation (inner folds select the penalty), within pool (z-scored per time control, over-the-board play its own pool). The elite case study reports the standardised axes for the six broadcast players with the most analysed decisions.	554 555 556 557 558 559 560 561 562
<i>Model comparison and the registered differential.</i>	563
State-only ($\beta=0$), search-only ($\alpha=0$), and fusion predictors are fit under one player split and compared on held-out cross-entropy and top-1 match. The registered hypothesis is an interaction: the fusion's gain over state-only is larger in high-swing than low-swing positions and in classical than blitz. It is tested with a player-clustered bootstrap and a linear mixed-effects model with player random intercepts.	564 565 566 567 568 569
<i>Convergent and recovery validation.</i>	570
Convergent: refit with the clock feature zeroed, then correlate held-out $\hat{d}$ with observed time-on-move, by game phase. Recovery: agents play a soft-max ($\beta_{\text{gen}}=6$) over fixed-depth win probabilities; their games pass through the identical pipeline, and a per-agent flat-prior posterior over depth (search likelihood, with the population β fit) is compared with the planted depth.	571 572 573 574 575 576
<i>Pre-registered replication programme.</i>	577
Hypotheses, pipeline, sampling seed, and the decision rule (success iff all three primary hypotheses replicate directionally with stated intervals) were registered before analysing each replication month (first registration: OSF KB4ZQ; corrected and prospective second registration: OSF A6NYK, whose primary month 2026-06 was registered before Lichess published the data). Each month was processed exactly once by the frozen pipeline. Deviations are disclosed in Results: the first registration's blitz prediction was too strong and its swing-direction convention reversed; the second registration's blitz-null prediction proved too weak in turn (blitz positive in the prospective month). Neither deviation touches the primary hypotheses.	578 579 580 581 582 583 584 585 586 587 588
<i>Reproducibility.</i>	589
Stockfish 18 (shared NNUE), Threads 1 per process across many single-thread engines, Hash 256 MB, $D=21$, $K=9$, 2×10^8 -node cap, driven through the <code>python-chess</code> UCI bridge (v1.11), which also provides all board and move representation; Maia-3 (maia3-79m) [29]; depth grid $\{2, 4, \dots, 22\}$; sampling seed 17 and 60 games per (rating band $\times$ time class) cell in every month; train/test split by player; model fits seeded. All analyses ran on a single dual-socket 64-core server with one GPU; the engine stage dominates the compute at roughly 1,400 core-hours per month of data, and all model fitting and analysis complete within about an hour. Strata and the primary interaction were fixed before held-out evaluation; post-hoc choices are labelled exploratory.	590 591 592 593 594 595 596 597 598 599

599 ***Ethics.***

600 This study is a secondary analysis of publicly available, pseudonymous game records;
601 no participants were recruited, no interaction or intervention took place, and no private
602 or identifiable personal information was collected or processed. Online players appear
603 only under their self-chosen public usernames; the named elite players analysed in
604 the individual profiles are public figures whose over-the-board tournament games are
605 published as a matter of course. As research limited to secondary analysis of public
606 data, the study did not require prior ethics approval and no formal ethics review was
607 sought; informed consent was not required for the same reason.

608
609 ***Data availability.***

610 All game records derive from public sources: the Lichess open database (<https://database.lichess.org>;
611 months 2025-09, 2026-04, 2026-05, 2026-06), openly licensed
612 Lichess broadcast exports of over-the-board play, and public games retrieved via
613 the chess.com Published-Data API (<https://api.chess.com>). The derived analysis
614 datasets—per-position engine depth trajectories, Maia-3 policies, and the model ten-
615 sors behind every result, for the primary month and all three replication months—are
616 deposited in the study’s Open Science Framework project (<https://osf.io/ruhy8>) under
617 a CC-BY licence; the deposit DOI will be minted on publication.

618
619
620 ***Code availability.***

621 The full pipeline (engine driver, dataset builder, model, confirmatory replication
622 script, and all figure code) and configuration files are openly available under the MIT
623 licence at <https://github.com/tamalrkm/depth-of-satisficing>. The pre-registrations
624 are archived in the same OSF project (registrations [10.17605/OSF.IO/KB4ZQ](https://osf.io/KB4ZQ) and
625 [10.17605/OSF.IO/A6NYK](https://osf.io/A6NYK)). Stockfish 18 and Maia-3 are third-party open-source
626 software.

627
628 ***Acknowledgements.***

629 The authors gratefully acknowledge the support from the Department of Science and
630 Technology (DST), Government of India, through the DST-FIST grant [Sanction No.:
631 SR/FST/MS-I/2022/116].

632
633
634 ***Author contributions.***

635 T.M. conceived the study, designed and implemented the latent-depth model and the
636 full analysis pipeline, ran the experiments, pre-registered and ran the replication pro-
637 gramme, and drafted the manuscript. K.W.R. contributed the intrinsic-performance-
638 rating and move-swing framework underlying the construct, advised on chess-cognition
639 methodology and validation design, and critically reviewed the manuscript. Both
640 authors approved the final version.

641
642 ***Competing interests.***

643 The authors declare no competing interests.
644

Ethics. 645
 The study uses only publicly released, openly licensed records of voluntary online and 646
 broadcast play; it involves no intervention and no private or personally identifying 647
 data beyond the public usernames and player names attached to those records, and 648
 so did not require ethical approval. 649
 650
 651
 652

References 653

- 654
 655
 656
 657
 658
 659
 660
 661
 662
 663
 664
 665
 666
 667
 668
 669
 670
 671
 672
 673
 674
 675
 676
 677
 678
 679
 680
 681
 682
 683
 684
 685
 686
 687
 688
 689
 690
- [1] Simon, H. A. A behavioral model of rational choice. *The Quarterly Journal of Economics* **69**, 99–118 (1955).
 - [2] Simon, H. A. Rational choice and the structure of the environment. *Psychological Review* **63**, 129–138 (1956).
 - [3] Gigerenzer, G. & Goldstein, D. G. Reasoning the fast and frugal way: Models of bounded rationality. *Psychological Review* **103**, 650–669 (1996).
 - [4] Russell, S. & Wefald, E. Principles of metareasoning. *Artificial Intelligence* **49**, 361–395 (1991).
 - [5] Griffiths, T. L., Lieder, F. & Goodman, N. D. Rational use of cognitive resources: Levels of analysis between the computational and the algorithmic. *Topics in Cognitive Science* **7**, 217–229 (2015).
 - [6] Lieder, F. & Griffiths, T. L. Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences* **43**, e1 (2020).
 - [7] Callaway, F. *et al.* Rational use of cognitive resources in human planning. *Nature Human Behaviour* **6**, 1112–1125 (2022).
 - [8] Tversky, A. & Kahneman, D. Judgment under uncertainty: Heuristics and biases. *Science* **185**, 1124–1131 (1974).
 - [9] Kahneman, D. & Tversky, A. Prospect theory: An analysis of decision under risk. *Econometrica* **47**, 263–291 (1979).
 - [10] Stahl, D. O. & Wilson, P. W. On players’ models of other players: Theory and experimental evidence. *Games and Economic Behavior* **10**, 218–254 (1995).
 - [11] Nagel, R. Unraveling in guessing games: An experimental study. *The American Economic Review* **85**, 1313–1326 (1995).
 - [12] Costa-Gomes, M., Crawford, V. P. & Broseta, B. Cognition and behavior in normal-form games: An experimental study. *Econometrica* **69**, 1193–1235 (2001).

- 691 [13] Camerer, C. F., Ho, T.-H. & Chong, J.-K. A cognitive hierarchy model of games.
692 *The Quarterly Journal of Economics* **119**, 861–898 (2004).
693
- 694 [14] Costa-Gomes, M. A. & Crawford, V. P. Cognition and behavior in two-person
695 guessing games: An experimental study. *The American Economic Review* **96**,
696 1737–1768 (2006).
697
- 698 [15] Crawford, V. P., Costa-Gomes, M. A. & Iriberri, N. Structural models of
699 nonequilibrium strategic thinking: Theory, evidence, and applications. *Journal*
700 *of Economic Literature* **51**, 5–62 (2013).
701
- 702 [16] Strzalecki, T. Depth of reasoning and higher order beliefs. *Journal of Economic*
703 *Behavior & Organization* **108**, 108–122 (2014).
704
- 705 [17] van Opheusden, B. *et al.* Expertise increases planning depth in human gameplay.
706 *Nature* **618**, 1000–1005 (2023).
707
- 708 [18] de Groot, A. D. *Thought and Choice in Chess* (Mouton, The Hague, 1965).
709
- 710 [19] Newell, A. & Simon, H. A. *Human Problem Solving* (Prentice-Hall, Englewood
711 Cliffs, NJ, 1972).
712
- 713 [20] Chase, W. G. & Simon, H. A. Perception in chess. *Cognitive Psychology* **4**, 55–81
714 (1973).
715
- 716 [21] Gobet, F. & Simon, H. A. Templates in chess memory: A mechanism for recalling
717 several boards. *Cognitive Psychology* **31**, 1–40 (1996).
718
- 719 [22] Ericsson, K. A., Krampe, R. T. & Tesch-Römer, C. The role of deliberate practice
720 in the acquisition of expert performance. *Psychological Review* **100**, 363–406
721 (1993).
722
- 723 [23] Campbell, M., Hoane Jr., A. J. & Hsu, F.-h. Deep Blue. *Artificial Intelligence*
724 **134**, 57–83 (2002).
725
- 726 [24] Silver, D. *et al.* A general reinforcement learning algorithm that masters chess,
727 shogi, and Go through self-play. *Science* **362**, 1140–1144 (2018).
728
- 729 [25] Regan, K. W. & Haworth, G. M. *Intrinsic chess ratings*. In *Proceedings of the*
730 *25th AAAI Conference on Artificial Intelligence*, 834–839 (2011).
731
- 732 [26] Biswas, T. T. & Regan, K. W. *Measuring level-k reasoning, satisficing, and human*
733 *error in game-play data*. In *2015 IEEE 14th International Conference on Machine*
734 *Learning and Applications (ICMLA)*, 941–947 (2015).
735
- 736 [27] McIlroy-Young, R., Sen, S., Kleinberg, J. & Anderson, A. *Aligning superhuman*
AI with human behavior: Chess as a model system. In *Proceedings of the 26th*
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD),

1677–1687 (2020).	737
	738
[28] Nosek, B. A., Ebersole, C. R., DeHaven, A. C. & Mellor, D. T. The preregistration revolution. <i>Proceedings of the National Academy of Sciences</i> 115 , 2600–2606 (2018).	739
	740
	741
	742
[29] Monroe, D., Eilender, G., Chalmers, P., Tang, Z. & Anderson, A. <i>Chessformer: A unified architecture for chess modeling</i> . In <i>International Conference on Learning Representations (ICLR)</i> (2026). arXiv:2605.19091.	743
	744
	745
	746
[30] Ruoss, A. <i>et al.</i> <i>Amortized planning with large-scale transformers: A case study on chess</i> . In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , Vol. 37, 65765–65790 (2024). arXiv:2402.04494.	747
	748
	749
[31] Samejima, F. Estimation of latent ability using a response pattern of graded scores. <i>Psychometrika Monograph Supplement</i> 34 , 1–97 (1969).	750
	751
	752
[32] Fischer, G. H. The linear logistic test model as an instrument in educational research. <i>Acta Psychologica</i> 37 , 359–374 (1973).	753
	754
	755
[33] Elo, A. E. <i>The Rating of Chess Players, Past and Present</i> (Arco, New York, 1978).	756
	757
	758
[34] Rasch, G. <i>Probabilistic Models for Some Intelligence and Attainment Tests</i> (Danish Institute for Educational Research, Copenhagen, 1960).	759
	760
	761
[35] Embretson, S. E. & Reise, S. P. <i>Item Response Theory for Psychologists</i> (Lawrence Erlbaum Associates, Mahwah, NJ, 2000).	762
	763
	764
[36] Daw, N. D., Niv, Y. & Dayan, P. Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. <i>Nature Neuroscience</i> 8 , 1704–1711 (2005).	765
	766
	767
	768
[37] Huys, Q. J. M. <i>et al.</i> Bonsai trees in your head: How the Pavlovian system sculpts goal-directed choices by pruning decision trees. <i>PLoS Computational Biology</i> 8 , e1002410 (2012).	769
	770
	771
[38] Ratcliff, R. & McKoon, G. The diffusion decision model: Theory and data for two-choice decision tasks. <i>Neural Computation</i> 20 , 873–922 (2008).	772
	773
	774
[39] Bilalić, M., McLeod, P. & Gobet, F. Inflexibility of experts—Reality or myth? Quantifying the Einstellung effect in chess masters. <i>Cognitive Psychology</i> 56 , 73–102 (2008).	775
	776
	777
	778
[40] Efron, B. Bootstrap methods: Another look at the jackknife. <i>The Annals of Statistics</i> 7 , 1–26 (1979).	779
	780
	781
	782